

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Information Technology) SEMESTER - 8 Summer 2025 (Regular)

Course :Bachelor of Technology (Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTITC801 - INTERNET OF THINGS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which of the following best defines Fog Computing?
 - A. Data processed only in centralized data centers
 - B. Processing and storage close to the data source, reducing latency
 - C. Cloud computing for mobile devices only
 - D. Storage of data in wearable devices
- 2 In a Wireless Sensor Network (WSN), which topology is commonly used for high fault tolerance?
 - A. Star
 - B. Ring
 - C. Tree
 - D. Mesh
- 3 Which of the following technologies is part of enabling IoT technologies and supports short-range low-power communication?
 - A. LTE-A
 - B. Bluetooth Low Energy (BLE)
 - C. 5G
 - D. Ethernet
- 4 Which of the following is a transport layer protocol commonly used with MQTT?
 - A. UDP
 - B. TCP
 - C. HTTP
 - D. IPX
- 5 Which of the following protocols is designed for resource-constrained IoT devices and supports RESTful interaction over UDP?
 - A. SOAP
 - B. MQTT
 - C. CoAP
 - D. XMPP

- 6 In the IoT Communication Network Layer, which sublayer manages the physical access to the network?
 - A. Network Transport Sublayer
 - B. Access Network Sublayer
 - C. IoT Network Management Sublayer
 - D. Gateway Sublayer
- 7 Which of the following platforms is a cloud-based solution provided by Amazon for building IoT applications?
 - A. Firebase
 - B. AWS IoT Core
 - C. IBM Watson
 - D. Azure IoT Hub
- 8 Which microcontroller board is commonly used in IoT for its built-in Wi-Fi and low power consumption features?
 - A. Arduino Nano
 - B. BeagleBone Black
 - C. Particle Photon
 - D. ESP32
- 9 Which of the following IoT hardware platforms is known for being credit-card-sized and supports GPIO, HDMI, and Linux OS?
 - A. Arduino Uno
 - B. Raspberry Pi
 - C. Cloudbit
 - D. ESP32
- 10 What is the purpose of 'Structural Health Monitoring' in Smart Cities?
 - A. Monitor energy usage in buildings
 - B. Detect structural damage in infrastructure like bridges and buildings
 - C. Improve street lighting efficiency
 - D. Enhance urban water supply systems
- 11 In the energy domain, which IoT application helps balance electricity supply and demand?
 - A. Renewable Energy Systems
 - B. Smart Vending Machines
 - C. Smart Grids
 - D. Weather Monitoring
- 12 What is the main IoT application used in retail for monitoring product availability?
 - A. Smart Payments
 - B. Inventory Management
 - C. Shipment Monitoring
 - D. Route Scheduling

Q2. Solve the following.

- A) Explain IOTWF Standardized Architecture in detail. 6
- B) Write Short note on : Edge Computing 6

Q3. Solve the following.

- A) List and Explain various types of Sensors and Actuators. 6

- B) Explain following access technologies with application area of each : 6
 (i) Zigbee (ii) NFC (iii) RFID (iv) MEMS
- Q4. Solve Any Two of the following.
- A) Write Short Note on : SCADA 6
 B) Describe different layers in Core IoT Functional Stack. 6
 C) Compare & Contrast : CoAP & MQTT 6
- Q5. Solve Any Two of the following.
- A) Compare Arduino & Raspberry Pi. 6
 B) Explain an IoT Software Platforms – REST and JSON-LD. 6
 C) Explain ESP32 with its characteristics. 6
- Q6. Solve Any Two of the following.
- A) Explain Home automation system in detail. 6
 B) Write Short Note on : Smart Agriculture 6
 C) Explain IOT Applications for Environment. 6

***** End *****